Pratik Adhikari

Lalitpur, Nepal
adhikaripratik2003@gmail.com | +977-9824163232
LinkedIn | GitHub | Portfolio

Research Interests

Renewable-energy integration, power-system monitoring and digital twins, and data-driven forecasting for grid modernization.

Education

Institute of Engineering, Pulchowk Campus Apr 2022 – Jul 2026
Bachelor of Engineering (BE), Electrical Engineering — Tribhuvan University, Lalitpur, Nepal

- Aggregate: 84.72% — Distinction (institutional scale; 80%+ is the Distinction threshold).
- Undergraduate thesis: *Digital Twin for Real-Time Monitoring and Short-Term Forecasting of a Grid-Connected Photovoltaic System*; advisor: [Prof./Assoc. Prof. Full Name].
- Selected coursework: Power System Analysis I & II, Power Electronics, Control Systems, Switchgear & Protection, Instrumentation I & II, Power Plant Equipment, Utilization of Electrical Energy, Engineering Mathematics I–IV, Computer Programming.

SOS Hermann Gmeiner School, Gandaki 2019 – 2021
Higher Secondary (Grade 11–12), Science — GPA: 3.64 / 4.00

Research Experience

Deadbeat Predictive Current Control with SVPWM and LCL Filtering for Grid-Connected Voltage-Source Converters 2026
Research Project, Department of Electrical Engineering, Pulchowk Campus

- Developed and analyzed a deadbeat predictive current controller with space-vector PWM (SVPWM) for a three-phase grid-connected voltage-source converter, replacing single-vector FCS-MPC to achieve fixed 10 kHz switching operation.
- Derived the continuous- and discrete-time dq-frame converter model and implemented a detuned deadbeat control law with capacitor-current-feedback active damping for an LCL-filtered grid interface.
- Designed and evaluated a 5 kW, 400 V, 50 Hz grid-connected VSC in MATLAB/Simulink, including L/LCL filter comparison, resonance analysis, harmonic-spectrum evaluation, THD/TDD analysis, and P–Q dynamic response.
- Demonstrated reduction in broadband current THD from 12.8% with baseline FCS-MPC + L filtering to 3.9% with the proposed deadbeat-SVPWM + LCL active-damping configuration under matched 10 kHz average switching frequency.
- Investigated LCL resonance, digital-delay effects in capacitor-current active damping, and deadbeat-controller de-tuning as practical stability considerations in predictive current control.
- **Methods and tools:** MATLAB, Simulink, predictive control, dq-frame modeling, SVPWM, LCL filter design, active damping, harmonic analysis.

Digital Twin for Real-Time Monitoring & Short-Term Forecasting of a Grid-Connected PV System 2025 – 2026
Undergraduate thesis, Department of Electrical Engineering, Pulchowk Campus

- Developed a physics-based MATLAB/Simulink digital twin of a grid-connected PV system (335 W module, boost converter, P&O MPPT, single-phase inverter) and validated it against field measurements, achieving $R^2 = 0.918$ on measured DC power.
- Designed and calibrated a low-cost LDR-based pyranometer ($R^2 = 0.974$ against a reference instrument), reducing irradiance RMSE by roughly 3× relative to satellite-derived data for site-specific validation.

- Built a SARIMA$(1,1,1)(1,1,1)_{24}$ pipeline for day-ahead irradiance and temperature forecasting ($R^2 = 0.921$ and 0.964), producing end-to-end power forecasts within 0.5% daily energy error.
- Coupled Arduino-based data acquisition to the Simulink model over serial communication, enabling live closed-loop digital-twin operation and real-time power prediction.
- Characterised asymmetric error behaviour across irradiance bands, isolating low-irradiance bias as the dominant residual and identifying targeted sensor and model refinements.
- **Methods and tools:** MATLAB, Simulink, Arduino, Python, SARIMA time-series modelling, sensor calibration, field measurement campaign.

Professional Experience

DeltaQ Engineering — Electrical Engineer (Remote, MEP) May 2026 – Present
California-based MEP engineering consultancy

- Perform electrical load calculations and system analysis for residential, multi-family, and commercial buildings, and prepare design documentation to applicable US codes and standards (NEC).
- Produce technical drawings, specifications, and coordination layouts in AutoCAD with senior engineers, and evaluate electrical equipment selections against project and code requirements.

Technical Projects

Smart Fertigation System [2023]

- Built an Arduino-based monitoring and control system (DHT-11, soil moisture sensors) that automates irrigation with prototype fertilisation control to improve water and nutrient efficiency.

Technical Skills

Programming: C/C++, Python, MATLAB scripting
Modelling & Simulation: MATLAB, Simulink, COMSOL Multiphysics, Proteus
Power Systems: DIgSILENT PowerFactory (coursework), AutoCAD Electrical, PCB design, protection coordination fundamentals
Data & Machine Learning: NumPy, Pandas, scikit-learn, statsmodels (SARIMA / time-series forecasting), Matplotlib
Instrumentation: Arduino, sensor calibration, serial data acquisition, field measurement campaigns
Other: LaTeX, Git, Bluebeam Revu

Teaching, Mentoring & Outreach

BloomED — STEM Content Creator (Volunteer) Jun 2021 – Jan 2022

- Produced simplified learning material explaining technical and scientific concepts for underprivileged students, developing science-communication and teaching skills.

Professional Development

- Supervised Machine Learning: Regression and Classification — DeepLearning.AI / Stanford, Coursera
- Supervised Learning with scikit-learn and Intermediate Python — DataCamp
- AutoCAD Electrical Essentials — SourceCAD

Languages

Nepali (native); English (professional working proficiency); Hindi (conversational)